

A Critical Analysis of the Prevalence of Technology-Facilitated Abuse in US College Students

Naman Gupta

n@cs.wisc.edu

Department of Computer Sciences
University of Wisconsin-Madison
Madison, Wisconsin, USA

Kate Walsh

klwalsh2@wisc.edu

Department of Psychology and Gender & Women Studies
University of Wisconsin-Madison
Madison, Wisconsin, USA

Sanchari Das

sanchari.das@du.edu

Ritchie School of Computer Science and Engineering
University of Denver
Denver, Colorado, USA

Rahul Chatterjee

rahul.chatterjee@wisc.edu

Department of Computer Sciences
University of Wisconsin-Madison
Madison, Wisconsin, USA

ABSTRACT

The ubiquitous use of technology by college students makes them vulnerable to harassment, harm, and intimidation via technological means. We evaluate the prevalence of such technology-facilitated abuse (TFA) among college students in the USA using a critical, feminist, and trauma-informed lens, which is essential to inform policymakers and advocates who support students. We surveyed 776 college students in a large R1 university located in the Midwest region of the USA to examine the prevalence of TFA faced by students marginalized by socio-economic factors, the support sought by student survivors, and the efficacy of support structures. Our findings indicate that 70% students experience TFA, but more than half of them do not seek support. Among the survivors who seek support, 93% students solely rely on informal resources like friends and family, and 6% solely seek support from formal networks such as survivor services or law enforcement. Therefore, we call on policymakers to direct attention to TFA, create tailored interventions to support marginalized students and propose campus-wide campaigns to spread awareness among college students.

CCS CONCEPTS

• **Security and privacy** → **Social aspects of security and privacy**; **Usability in security and privacy**; • **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

critical race theory; technology abuse; domestic violence; intimate partner violence; technology-facilitated abuse; critical; college populations; trauma; trauma-informed computing; transgender; gender-based violence; computer security and privacy

ACM Reference Format:

Naman Gupta, Sanchari Das, Kate Walsh, and Rahul Chatterjee. 2024. A Critical Analysis of the Prevalence of Technology-Facilitated Abuse in US College Students. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3613905.3652036>

Content Warning. This paper contains descriptions of intimate partner violence, digital abuse, online harassment, racial violence, and mental health topics including trauma.

1 INTRODUCTION

Today, college students rely heavily on technology for various aspects of their lives, including learning, social collaboration, financial management, entertainment, health, and social interaction. However, the widespread use of technology also exposes students to increased safety risks and leaves them vulnerable to technology-facilitated abuse (TFA). This form of abuse encompasses behaviors such as hate speech, spying, stalking, harassment, doxxing, and bullying targeted at individuals. College students may overuse and abuse technology [72, 75], face harassment on social media [41, 80], rely on technology for dating [31], and share intimate images for “sexting” [50, 65, 106, 107, 109, 114].

College communities are complex social systems in which students are immersed in novel environments, engaging with individuals of diverse backgrounds. Various power structures manifest within these contexts, including fellow students, instructors, and staff members [8, 75]. The students may find restricted availability of family and community support with high levels of academic stress and mental health concerns that may expose them to TFA [8, 64]. Marginalized identities of students based on gender, sexual orientation, relationship status, race/ethnicity, disability, income status, graduate/undergraduate, may compound their risk of interpersonal violence [21, 94, 96, 142].

Prior studies [33, 43, 85] observed that college students experience TFA from an intimate partner (TFA-IPV). While previous studies highlight the importance of examining TFA victimization on college campuses, they have several limitations. For example, these studies predominantly select students facing TFA from an intimate partner, overlooking abusive experiences from alternative

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3652036>

sources, and failing to account for crucial factors contributing to the marginalization. They have a binary perspective on Gender and sexual orientation (Male/Female) and conform to hetero-normative relationships. Moreover, they use a criminology lens using routine activity theory and lifestyle theory [49, 95, 99] to gauge pro-abuse behavior and attribute student's abuse experiences through their lifestyle instead of validating survivor experiences. Therefore, we use a critical [42, 142] and trauma-informed [39, 142] lens to center survivor experiences and value the mitigation strategies they adopt to cope with the abuse, such as support-seeking.

We investigate the following two research questions:

RQ1. How prevalent is TFA among college students, and how does it vary for students marginalized by socio-economic factors?

RQ2. Who do survivors of TFA ask for support regarding the abuse?

We conducted the first quantitative study to analyze the prevalence of TFA on college campuses and identify how marginalization such as gender, sexual orientation, relationship status, race/ethnicity, disability, income status, graduate/undergraduate, increases the risk of experiencing TFA through a Critical Race Theory (CRT) lens [42, 142]. We validate the survivor's experiences through a trauma-informed approach [39, 142] and identify the support systems students reach out to after facing TFA.

We conducted an online survey study with students in a large R1 university located in the Midwest region of the USA.¹ We found that Cis-Women are 1.8 times and LGBTQ+ students are 2.0 times more likely to face TFA than Cis-Men. Furthermore, students who reported having a disability are nearly three times (3.0x) as likely as students without a disability to face TFA. The most common TFA experience faced by 472 (60%) students was unwanted contact through calls, emails, voicemails, texts, or instant messages. In addition, 402 (51%) students faced image-based TFA experiences such as non-consensual posting or sharing unwanted messages, pictures, and videos online. However, we found that more than half (53%) of the survivors of TFA do not seek support, 93% solely seek support from informal support structures, and merely 6% seek support from formal resources. We observed that while Cis-Women and LGBTQ+ seek support from informal support structures, Cis-Men are less likely to seek any support after facing TFA.

Therefore, we discuss the findings to inform the implications for future research practices, design of technology, and college campus interventions.

2 BACKGROUND

Technology-Facilitated Abuse (TFA) includes online abuse, in-person spying, and harassment and violence. Online forms include cyber-bullying on social media or posting unwanted and intimate messages, pictures, or videos online [50, 109, 114, 130]. In-person abuse can look like tracking location without consent, unwanted contact via messages, hidden spyware and stalkerware on survivor's smartphone, re-purposed dual-use applications [9, 18, 22, 38, 53, 88, 100, 112, 116, 132, 146]. Further, an abuser could impersonate survivors' social media accounts, change passwords or recover questions to lock them out of their accounts, share intimate images or messages without their consent, repurpose shared devices

like computers, phones, iPads, cameras, smart thermostats, etc. to threaten the life or safety of their friends, family, or pets [37, 127], and control or restrict access to finances, financial services, or financial statements [20, 27, 102, 138, 139]. Finally, due to technological advancements, adoption, and increased use of smart-home technology [120, 126, 127], the prevalence of TFA has worsened, especially during the COVID-19 pandemic, showing a dire need for interventions and mitigations [12, 16, 58, 59, 82–84, 101, 121, 133, 136].

Due to the invisible nature of TFA, survivors may not be adequately equipped to confront the situation, as they may lack the necessary knowledge to mitigate the abuse [54, 88]. Several studies [52, 63, 134] proposed an intervention called Tech Clinic where technical experts address technological concerns, provide emotional support and help survivors with a safety plan [39, 123, 135]. However, college students may not have these services available on campus or might not be aware of the resources in the community as they exist only in 3 cities in the US. Moreover, students from marginalized communities are discriminated against, may face exclusionary experiences, and have more difficulty accessing support services [124].

Furthermore, prior research [40, 45, 74, 137, 142] suggests that survivors may reach out to their friends and family to receive support to cope with distress. The availability of social support reduces the adverse outcomes of abuse and helps them leave an abusive relationship [149]. However, not all survivors seek help and try to mitigate the abuse by themselves with assistance from online resources [79, 88, 142]. Therefore, it is important to understand the form of support available (or lack thereof) to design interventions, especially in the absence of tech experts.

Research gap. While previous studies highlight the importance of examining TFA victimization on college campuses, they lack in various aspects. First, these studies predominantly select female students in hetero-normative relationships and overlook online abuse from strangers. Second, prior works exclusively use a criminology lens using routine activity theory and lifestyle theory [49, 95, 99] to find justifications for student's abuse experiences through their lifestyle [33, 43, 85]. Thus we lack an inclusive understanding of TFA among college campuses and how marginalized students are affected by TFA.

We therefore center survivors' experiences through a trauma-informed lens [39, 142] and analyze the marginalization faced by students through a critical lens [42, 142].

3 METHODS

3.1 Recruitment

We conducted an online survey with students at a large R1 university located in the Midwest region of the USA. The survey² was titled "Sexual and Emotional Experiences Among University Students." An email invitation with a unique survey link was sent to a random sample of 4,000 students. A total of 844 students began the survey; however, only 802 students responded to the attention check items appropriately. We removed 14 participants because they reported that the experiences listed as TFA were not alarming

¹The university has a White-dominated student population.

²The survey is available at this link: https://drive.google.com/file/d/1p9_Kj9d_1Ahu28edyrg_OucDFHWA9b86/view

or spying-related. We acknowledge that quantitative surveys have inevitable interpretation biases, especially the way the experiences were interpreted by the participants. We plan to modify the questions in our future work to minimize erroneous interpretations of abuse (Section 5).

In total, 776 students were considered for data analysis ($mean = 21.12$, $median = 20$, $range = [18, 36]$). We report the demographic distribution of the participants in Table 1. Due to the small sample size, some categories were collapsed for analysis. Since TFA is an example of gender-based violence, we observe that the demographic distribution of our sample is representative of the demographic of a large R1 university located in the Midwest region of the USA.³ Among the sample, 27% identified as Cis-Men, 44% as Cis-Women, and 27% as LGBTQ+.⁴ Most students (40%) were single, while 36% students were in a committed monogamous relationship, and 18% students were involved in an ongoing hook-up, friends with benefits, were “talking” to someone, polyamorous, etc. (which we refer to as “non-exclusive relationship”). A small fraction of the students (4%) were married, divorced, or widowed. Among the participants, 68% identified as White, 31% identified as belonging to Black, Asian, Indian or Alaska Native, Native Hawaiian or Pacific Islander, or Multi-racial and people of color (BIPOC). In terms of ethnicity, only 11% identified as Hispanic. Most students (81%) in our survey are undergraduates, but 18% are graduate students. Furthermore, 20% students identified as first-generation college students and 15% as having a disability including intellectual disability, learning disability, sensory disability, mobility disability, mental illness, disability. Approximately 51% students self-reported being low-income (participants who (1) receive Pell Grant, (2) Financial Aid, or (3) have difficulty meeting basic needs “half the time, most of the time, or always”).

3.2 Ethical Considerations and Positionality.

Informed consent was obtained from all participants, and they could skip any question or withdraw from the study at any time. Some participants may feel distressed or uncomfortable when answering questions about prior traumatic experiences like sexual violence and intimate partner violence. The three authors of this paper are trained advocates who support survivors in various capacities by applying trauma-informed principles of care. If participants experience psychological stress or discomfort, one member is a licensed clinical psychologist who could respond to distress and provide referrals to the participants. All participants receive their contact information and confidential survivor services resources. The responses were stored in a secured Google Drive folder, accessible only to approved study personnel. We anonymize any identifiable information about the participants. The participant’s email were separately stored for payment purposes. We did not mention the study title in the payment, and the consent document given to the participant was limited to the amount necessary to achieve the research aims.

³The university has a White-dominated student population.

⁴Gender identity expressed as Non-binary, Two-spirit, Gender-Fluid and Gender-Nonconforming and Sexual Orientation expressed as Lesbian, Gay, Bisexual, Transgender, Queer, Asexual, Pansexual

3.3 Data Analysis

We analyzed the responses from the participants by exporting the CSV file responses from Qualtrics. Then, we cleaned the CSV file and removed the responses that failed the attention checks. We used Python to clean the data, and R to perform the statistical tests.

We used logistic regression to observe the association between demographics such as gender, sexual orientation, relationship status, race/ethnicity, disability, income status, graduate/undergraduate, and TFA or TFA-IPV. We measure the interaction of Gender and Race factors by controlling for other factors in multivariate models with TFA and TFA-IPV as the outcome. We report the Odds Ratio (OR) and 95% confidence interval (CI) of the logistic regression (Table 2). Using a Critical Race Theory (CRT) lens [42], we hypothesized that marginalized groups would be more likely to report TFA, therefore, across each demographic, the most privileged group served as the reference category (e.g., white, Cis-Men, non-Hispanic, single, non-low-income, non-disabled, non-first generation, etc.).

4 RESULTS

4.1 RQ1: Prevalence of Technology-Facilitated Abuse

We corroborate the prevalence of Technology-Facilitated Abuse (TFA) in college students with prior works and use a critical and trauma-informed lens for the analysis [39, 78]. In total, 545 (70%) students reported facing at least one form of TFA, while 161 (29%) students faced TFA from an intimate partner (TFA-IPV).

To validate the prevalence of TFA, we compare the associations of demographics with Sexual Violence (SV) (638 (82%)) and Sexual Violence by an intimate partner (SV-IPV) (73 (9%)). SV consists of scenarios of unwanted sexual contact, including non-consensual oral, anal, vaginal, or penetrative sex and touching, kissing, fondling, or grabbing sexually. SV-IPV consists of experiences of sexual violence from a current or former romantic partner. We observe that the prevalence rates and demographic associations with TFA and TFA-IPV are similar to those for SV and SV-IPV.

Gender: We observed that Gender has a strong association with TFA-victimization ($\chi^2 = 33.979$, $p = 0.000006$, $df = 6$). Cis-Women ($odds = 1.833$, $p = 0.0102$, $CI = [0.986, 2.680]$) are 1.833 times and LGBTQ+ students ($odds = 1.951$, $p = 0.0202$, $CI = [0.850, 3.051]$) are 1.951 times more likely to face TFA than Cis-Men. Although we did not observe a statistically significant result in the case of TFA faced from an intimate partner (TFA-IPV), 23% of LGBTQ+ participants and 23% of Cis-Women participants are susceptible to TFA-IPV, which is much higher than Cis-Men (13%). The prevalence statistics are consistent with the prior work, which shows that violence is a gendered issue [36]. Prior studies show that LGBTQ+ students are likely to face higher and more severe amounts of violence and harassment [28, 57, 96, 130, 140] as they face stigma around coming out, going through gender transition [47, 60, 104] and expressing their identity on Social media [24, 25, 35, 76, 119, 140]. Furthermore, reports suggest that “sextortion” scams on dating apps are becoming more common against LGBTQ+ communities [55]. Lack of family support [111] and a hostile political and legal environment can lead to their withdrawal from society and increase the risk of violence. Our results corroborate that Cis-Women and LGBTQ+

Table 1: Table shows: (1) the prevalence statistics of victimization to TFA and TFA-IPV grouped by the Demographic variables. Among 776 respondents, 545 reported experiencing TFA, and 161 experienced TFA from an intimate partner. (2) Help-seeking statistics of 545 students who experienced TFA, grouped by demographic variables. The percentage within parenthesis represent the percentage of participants in the column within the social demographic.

Social Demographic	Total participants (<i>n</i> = 776)	Victimization		Help-Seeking of 545 TFA survivors		
		TFA (<i>n</i> = 545)	TFA-IPV (<i>n</i> = 161)	No Help	Informal [Ⓢ]	Formal ^λ
Gender-Sexual Orientation						
Cis-Women	344 (44%)	256 (74%)	81 (23%)	121 (47%)	186 (72%)	10 (3%)
Cis-Men	216 (27%)	119 (55%)	29 (13%)	81 (68%)	35 (29%)	2 (1%)
LGBTQ+ [†]	216 (27%)	170 (78%)	51 (23%)	88 (51%)	82 (48%)	9 (5%)
Race						
White	532 (68%)	388 (72%)	117 (21%)	214 (55%)	172 (44%)	10 (2%)
BIPOC*	244 (31%)	157 (64%)	44 (18%)	76 (48%)	78 (49%)	11 (7%)
Ethnicity						
Hispanic	90 (11%)	62 (68%)	21 (23%)	35 (56%)	26 (41%)	2 (3%)
Non-Hispanic	686 (88%)	483 (70%)	140 (20%)	255 (52%)	224 (46%)	19 (3%)
Relationship Type						
Single	315 (40%)	213 (67%)	57 (18%)	120 (56%)	91 (42%)	9 (4%)
Committed monogamous relationship	281 (36%)	194 (69%)	61 (21%)	100 (51%)	94 (48%)	8 (4%)
Non-exclusive relationship*	143 (18%)	112 (78%)	32 (22%)	54 (48%)	57 (50%)	4 (3%)
Married, Divorced, or widowed	37 (4%)	26 (70%)	11 (29%)	16 (61%)	8 (30%)	0 (0%)
Graduate Student?						
Undergraduate student	632 (81%)	446 (70%)	126 (19%)	234 (52%)	210 (47%)	13 (2%)
Graduate student	144 (18%)	99 (68%)	35 (24%)	56 (56%)	40 (40%)	8 (8%)
Low Income						
Yes	401 (51%)	289 (72%)	94 (23%)	151 (52%)	137 (47%)	15 (5%)
No	375 (48%)	256 (68%)	67 (17%)	139 (54%)	113 (44%)	6 (2%)
Disability						
Yes	121 (15%)	105 (86%)	37 (30%)	54 (51%)	51 (48%)	5 (4%)
No	655 (84%)	440 (67%)	124 (18%)	236 (53%)	199 (45%)	16 (3%)
First Generation College Student						
Yes	160 (20%)	111 (69%)	38 (23%)	56 (50%)	54 (48%)	8 (7%)
No	616 (79%)	434 (70%)	123 (19%)	234 (53%)	196 (45%)	13 (2%)

[†] Gender identity expressed as Non-binary, Two-spirit, Gender-Fluid and Gender-Nonconforming and Sexual Orientation expressed as Lesbian, Gay, Bisexual, Transgender, Queer, Asexual, Pansexual * Black, Indigenous (American Indian or Alaska Native and Native Hawaiian or Pacific Islander), and People-of-Color (Asian and Multi-racial) ♣ Ongoing hookup, friends-with-benefits, “talking” to someone, or non-exclusive relationship, polyamorous
[Ⓢ] Informal support includes seeking help from close friends, roommates, family or relatives, or an intimate partner ^λ Formal survivor support service like University Health Services, a rape crisis center, a domestic violence organization, or a technology resource like the Tech Clinic or Geek Squad or law enforcement resource (e.g., Cybercrime Unit, police)

students experience significantly higher TFA rates compared to Cis-Men students.

Disability: We observe that disability (e.g., intellectual disability, learning disability, sensory disability, mobility disability, mental illness, etc.) was significantly associated with the TFA ($\chi^2 = 18.25$, $p = 0.0011$, $df = 4$). The students who reported having a disability are more than twice (2.8x) as likely as students without a disability to face TFA (*odds* = 2.778, $p = 0.005$, $CI = [0.235, 1.359]$). Previous works show that the lack of accessible technology could add additional privacy and safety risks to people with disabilities [4–6, 91, 141]. Furthermore, a “visible” vs. “invisible” disability could

affect how young adults use dating apps to reduce unnecessary connections [103, 105].

Race: Furthermore, the prevalence of TFA among BIPOC students is similar to that of white students in our sample. A possible reason could be that the majority of the students in the BIPOC group are Asian (142 (18%)), and only 25 (3%) are Black, 3 (0%) indigenous, and 74 (9%) multiracial. Since we grouped all these races into BIPOC for more statistical power, any differences between individual races may be lost. Future studies should over-sample students from these under-represented races to truly understand their marginalization, as we discuss in Section 5. Prior works show that Black women are 35% more likely to experience IPV than White women due

Table 2: The table shows odds ratio (OR) and 95% confidence intervals in brackets for the regression analysis of TFA-victimization.

	TFA	TFA-IPV	SV-IPV	SV
Year [Graduate]	0.994 (0.536, 1.452)	1.237 (0.619, 1.855)	1.598 (0.581, 2.615)	0.976 (0.445, 1.507)
Ethnicity [Hispanic]	0.870 (0.421, 1.320)	1.075 (0.479, 1.671)	0.825 (0.182, 1.468)	1.030 (0.375, 1.685)
Relationship [Married]	1.074 (0.148, 1.999)	1.528 (0.186, 2.870)	4.487 ^{***} (-0.459, 9.434)	1.123 (-0.008, 2.253)
Relationship [Monogamous]	1.030 (0.656, 1.404)	1.237 (0.722, 1.751)	2.908 ^{***} (0.936, 4.881)	1.549 [‡] (0.865, 2.232)
Relationship [Non-Exclusive]	1.420 (0.734, 2.105)	1.129 (0.560, 1.697)	3.307 ^{***} (0.841, 5.774)	1.486 (0.631, 2.341)
Low Income [Yes]	1.019 (0.677, 1.362)	1.244 (0.778, 1.710)	1.345 (0.634, 2.056)	1.037 (0.620, 1.454)
First Generation [Yes]	1.051 (0.607, 1.495)	1.181 (0.652, 1.710)	1.274 (0.510, 2.038)	1.084 (0.538, 1.631)
Disability [Yes]	2.411 ^{***} (1.030, 3.793)	1.573 [‡] (0.844, 2.303)	1.181 (0.420, 1.943)	2.241 ^{**} (0.593, 3.888)
Gender [Cis-Women]	1.833 ^{**} (0.986, 2.680)	1.379 (0.596, 2.162)	1.440 (0.230, 2.651)	2.673 ^{***} (1.160, 4.186)
Gender [LGBTQ+]	1.951 ^{**} (0.850, 3.051)	1.676 (0.615, 2.736)	2.167 [‡] (0.198, 4.135)	2.135 ^{**} (0.705, 3.564)
Race [BIPOC]	0.617 [‡] (0.271, 0.963)	0.572 (0.083, 1.060)	0.522 (-0.204, 1.247)	0.644 (0.255, 1.033)
Gender [Cis-Women] & Race [BIPOC]	1.319 (0.269, 2.368)	2.536 [‡] (-0.063, 5.135)	2.019 (-1.252, 5.291)	0.878 (0.072, 1.684)
Gender [LGBTQ+] & Race [BIPOC]	1.690 (0.192, 3.188)	0.960 (-0.106, 2.025)	1.447 (-0.923, 3.817)	1.223 (-0.010, 2.455)
Constant	1.384 (0.774, 1.995)	0.137 ^{***} (0.059, 0.215)	0.023 ^{***} (0.002, 0.045)	2.129 ^{***} (1.072, 3.185)
Log Likelihood	-446.907	-382.592	-224.345	-339.148
Akaike Inf. Crit.	921.813	793.184	476.691	706.296

*P-values:** $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

to systemic racism and white supremacy rooted in American history [1, 42]. Furthermore, Indigenous peoples (84.3% women & 81.6% men) face disproportionately high rates of IPV, especially from a non-indigenous partner, due to colonial and generational trauma [16, 44, 82].

Economic Status: We did not observe a statistically significant difference in marginalization due to low economic status or first-generation status. However, prior work suggests that survivors face economic insecurity and that violence can worsen their economic status [20, 81]. In particular, violence increases women's dependence on partners or families for necessities and reduces access to resources. The survivors may face problems such as utility disconnections, housing instability, food scarcity, and difficulty accessing medical care. Voth et al. observed a negative association between social support and economic hardship [139]. A mix of financial aid, advocacy, education, and support can alleviate economic distress. Future studies should investigate the relationship between economic status and TFA. We discuss the support-seeking dynamics of survivors from different economic and educational statuses in Section 4.2.

The abuse experiences shown in Table 3 may be more severe than others in the specific context in which they were experienced. Prior research demonstrates that repeat TFA-victimization (more than three times) is more dangerous in the "cycle of violence" [146]. The most common TFA experience faced by 472 (60%) students was unwanted calls, emails, voicemails, etc. In addition, 402 (51%) students faced image-based sexual abuse experiences such as non-consensual production and sharing of unwanted messages, pictures, and videos. Prior works by Henry and Powell et al. [65–69, 106, 107] show the prevalence of image-based sexual abuse (IBSA) that causes emotional distress to the survivors of IPV and dating violence. IBSA is disproportionately high among young women [109, 115, 150]. Moreover, minority ethnic populations and LGBTQ+ students are likelier to report a more significant negative impact on their health and relationships [67].

We recognize that the intersection of socio-economic demographics compounds vulnerability to digital safety and privacy risks [91, 92, 118, 125]. Therefore, an intersectional analysis is required to observe the interlocking axes of social, political, economic, and historical reasons for marginalization [21, 78, 142]. Due to the limitation of our current dataset, we cannot fully analyze these relationships due to lack of statistical power. We further expand the limitations of our study in Section 5.

4.2 RQ2: Support-Seeking Behavior of TFA survivors

We recognize the difference between disclosure and request for support, especially in different contexts of informal or formal support structures [48, 60, 128]. In the future, we could draw parallels from disclosures of abuse from research on IPV and Sexual Violence [19, 70, 128, 143, 144]. The students may have interpreted this question either way. In the context of this paper, we consider only the support-seeking behavior of the students. We analyze demographic differences through a multivariate logistic regression model (see Table 4).

Informal Support: We observe that more than half (290 (53%)) of the students who faced TFA do not seek any support. We observed that at least 500 (91%) students reported having a positive friend support network – measured using MPSS Scale [151] – but only half (250 (50%)) of them sought any informal support, such as friends (40%), roommates (15%), family and relatives (19%), or an intimate partner (4%). We observe that Cis-Men are 64.5% and 55.9% less likely to seek support after facing TFA than Cis-Women ($odds = 0.355$, $p = 0.000494$, $CI = [0.148, 0.562]$) and LGBTQ+ survivors ($odds = 0.441$, $p = 0.012800$, $CI = [0.157, 0.725]$) respectively. Moreover, we observed that Cis-Women and LGBTQ+ are $3.0\times$ ($odds = 2.964$, $p = 0.000304$, $CI = [1.217, 4.712]$) and $2.4\times$ ($odds = 2.432$, $p = 0.007495$, $CI = [0.848, 4.016]$) more likely to seek support from friends, family, relatives, or an intimate partner as compared to Cis-Men respectively. Ybarra et al. found that fewer

Table 3: Frequency distribution of victimization to abusive experiences (n = 776).

Item	At least Once	Once	Twice	3+ times
made unwanted phone calls, emails, voicemails, texts, or instant messages?	472 (60%)	133 (17%)	85 (10%)	254 (32%)
posted unwanted messages, pictures, or videos online?	204 (26%)	88 (11%)	54 (6%)	62 (7%)
shared intimate images or messages about you without your consent?	198 (25%)	98 (12%)	32 (4%)	68 (8%)
convinced you to share your account access, passwords, or recovery questions	88 (11%)	58 (7%)	15 (1%)	15 (1%)
Used technology to threaten the life or safety of your friends, family, or pets?	79 (10%)	52 (6%)	14 (1%)	13 (1%)
Installing spyware on your phone?	10 (1%)	7 (0%)	2 (0%)	1 (0%)
Using applications like GoogleMaps, iCloud, or Snapchat maps?	160 (20%)	50 (6%)	22 (2%)	88 (11%)
Using shared devices like computers, phones, iPads, cameras, smart-devices,	63 (8%)	26 (3%)	8 (1%)	29 (3%)
Controlled or restricted access to finances, financial services and statements	24 (3%)	14 (1%)	2 (0%)	8 (1%)

men (21%) sought support from friends and family than women (43%) and LGBTQ+ (48%) [150]. Prior literature shows that 81% of adult women who face physical or sexual IPV disclose to at least one informal support, while only 57% of men disclose abuse [10]. Cis-Men may face challenges in seeking support due to social stigma [110]. Moreover, we do not observe significant differences in marginalized students seeking informal support. However, prior work found that Black, Hispanic, or younger women with low-income status who face IPV have lower disclosure rates compared to older white women with high-income status [19].

Formal Support: Merely 21 (3%) of our participants sought formal support from survivor services, tech clinics, cybercrime unit of law enforcement or organizations such as Dean of Students Office, the Police, or someone in a position of power. This shows that although formal resources are available to students, they are underutilized. Previous work shows that students (especially undergraduates) might even be unaware of the resources [93], facing a problem of service discovery [145]. Woodlock et al. note that survivors of TFA are reluctant to approach the criminal justice system and law enforcement for fear of being dismissed, with institutional betrayal compounding their grief and trauma from disenfranchisement [148]. Marginalization through social demographics can add barriers to seeking support, as participants may receive misogynistic or racist remarks and non-helpful comments from formal structures [17, 30, 82, 117]. For brevity, we leave the analysis of the association of support-seeking behavior of students belonging to different demographics, such as gender, sexual orientation, relationship status, race/ethnicity, disability, income status, graduate/undergraduate, as part of future work.

Effectiveness of Support: Among those who sought support (n = 252), 166 Students (65%) found that the support was effective in fixing the technical problem they were facing. In contrast, 80 (31%) mentioned that the support did not fix the issue. Most students 93% said that the support made them feel better. Finally, only 9 (3%) reported that they were referred to formal services such as Tech Clinic, Geek Squad, or Cybercrime Unit. Therefore, we observe that even though survivors seek support from informal support structures, it may not fix the technical concerns, make them feel better, or connect them to formal services.

5 DISCUSSION AND IMPLICATIONS OF THE STUDY

We analyze the prevalence of technology-facilitated abuse in Mid-western U.S. college students through a trauma-informed and critical lens. We observe that students belonging to marginalized social demographics like LGBTQ+, disability, and low-economic conditions have a higher chance of facing TFA. Similar to implications discussed by Klein et al., we discuss implications for research, practice and technology design, and policy interventions in campus [77]. Furthermore, we recognize that the study is not without certain limitations and discuss future directions of research.

Implication for Research: We found that Cis-Women are 1.8 times and LGBTQ+ students are 2.0 times more likely to face TFA than Cis-Men. Furthermore, students who reported having a disability are nearly three times (3.0x) as likely as students without a disability to face TFA. Socioeconomic factors amplify the risks to digital safety and privacy [91, 92, 118, 125], demanding intersectional analysis to untangle the complex web of marginalization. However, the lack of representation of intersectional identities prevented us from performing such an analysis. Oversampling underrepresented groups and a qualitative approach allows a deeper understanding of how, say, students with low-economic conditions and disabilities face challenges while interfacing with technology can help us inform the design of technology and reduce friction. Additionally, we found that more than half (53%) of the survivors of TFA do not seek support, 93% solely seek support from informal support structures, and merely 6% seek support from formal resources. Therefore, establishing a timeline and context of abuse and the survivor's support-seeking behavior will highlight the temporality and severity of the abuse and actions taken to mitigate the abuse. The sooner survivors know about the support services, the more informed they are to navigate their abusive relationship. Their lived experience will help us understand what their support-seeking interactions look like.

Lastly, this study discussed the prevalence of TFA in the US college context. Future work requires further research to understand the prevalence of technology-facilitated abuse in young adult populations outside college campuses and countries outside the US. Recently, researchers have looked at the landscape of technology-facilitated abuse in tribal Australia [30, 61], women in the Global South (India, Pakistan, Bangladesh, and Singapore) [117] and have

highlighted the context of abuse when family dynamics and technology adoption are significantly different from North America. For example, Sambasivan et al. [117], observe that a fully-clothed photo, name, or phone number released publicly will be a dangerous issue for some participants with serious societal implications that may not be the case in American society. We hypothesize that the sociotechnical challenges in Global South countries may present insights that allow us to design the study appropriately. Therefore, exploring the landscape of technology-facilitated abuse in Rural communities can challenge the applicability of lessons learned in prior literature on urban cases of TFA.

Implication for Design of Technology: The most common TFA experience faced by 472 (60%) students was unwanted contact through calls, emails, voicemails, texts, or instant messages. In addition, 402 (51%) students faced image-based TFA experiences such as non-consensual posting or sharing unwanted messages, pictures, and videos online. The rising trend of Image-based Sexual Violence (IBSV), especially nonconsensual AI-generated deepfake pornography towards women and LGBTQ+, may worsen with the ease of accessibility to AI technologies nowadays [131]. Prior works indicate the need to empower young adults with control and agency through online safety features to recognize and deal with risks of online harassment [2, 3, 7, 11, 13–15, 23, 26, 51, 73, 87, 113]. Prior works show that teens want accountability and evidence-based mechanisms [129] from social media platforms for actions like reporting and ensuring their online safety [2]. To combat online harassment at its root, prior works call for community-engaged solutions that are preventative for the perpetrator (warnings, education, blocking, and punishment) instead of reactive design interventions [2, 3], and provide intelligent guidance such as safer responses and nudges for seeking support from family [2, 51, 71, 87, 108]. Moreover, in-person TFA experiences also require careful design interventions using technology through a trauma-informed lens [39, 51]. Explicit permission usage and privacy disclaimers in dual-use applications on the mobile operating system can be enabled against non-consensual location tracking [51, 90]. Prior works also show that the survivors of violence may use technology such as social media and instant messaging platforms to form support networks and resist violence [71, 108, 145].

Campus Interventions: Campus policies can be informed to focus attention on educating the students about the dangers of TFA. We observed that while Cis-Women and LGBTQ+ seek support from informal support structures, Cis-Men are less likely to seek any support after facing TFA. Further, only 3% sought help from formal resources. This requires a case for policy decisions to provide evidence-based guidance and formal services accessible to students. The policymakers should consult with practitioners such as forensic nurses, activists, social workers, legal advocates, and law enforcement [112, 123]. Understanding the prevalence can help policymakers, advocates, and educators focus attention on interventions and inform their designs to make university campuses safer. Recently, interventions have been conducted to support survivors of IPV and sexual violence through violence awareness campaigns, bystander interventions through student organizations, residential committees, and survivor services [32, 93]. In addition, mental

health counselors, education programs, and peer networks are introduced to new students. In a similar vein, Tech Clinic could be integrated with the IT department and survivor services in colleges to support the survivors. This will enable the IT departments to provide cybersecurity advice with empathy and avoid victim blaming [122]. A push for conversation around consent in the technology realm, relevant resources on TFA, culturally-attuned and trauma-informed support, and push for educational content around safety planning through technology that will empower marginalized students and build their confidence [56]. Munro et al. suggests a one-stop-shop strategy to provide information on all services that a survivor requires to cope with abuse [98]. Further, most universities' Title IX [34] and Diversity, Equity, and Inclusion (DEI) office can provide training to faculty/staff to support marginalized students who face disproportionate violence and hardship to help provide them with appropriate resources and academic accommodations.

Therefore, appropriate interventions are required to make support resources accessible and available, and designed for marginalized groups of students such as students with disability and lower socioeconomic status with financial aid.

Limitations and Future Work We discuss how the limitations of the study may have influenced the results and suggest areas for future research. We believe that students who had prior abusive experiences may be more willing to participate in the study to share their experiences. However, the invisibility and hidden nature of TFA make it challenging to discover, acknowledge the abuse, or ask for support [89]. Therefore, the prevalence statistics that we observed could likely be an underestimate of the actual prevalence of TFA among students and therefore make quantitative studies on abuse experiences challenging to conduct and analyze.

Moreover, some participants may not have considered scenarios as abusive or in the context of violence. For example, we asked the participants if they categorized "Has someone spied on you, monitored your activities online, or tracked your location?". The students may have been interpreted as non-threatening and non-abusive and could be labeled as benign use of technology. Moreover, prior works locate how TFA resides in the spectrum and continuum of abuse and violence [46, 62, 97, 102, 147]. Although the students resonated with TFA experiences, more work is required to understand where do the students place them on a scale of aggression and harm. Moreover, we plan to include more TFA experiences (e.g., looking through devices without permission, derogatory name-calling using technology [86]), and recognize that certain experiences could happen in non-abusive scenarios as well (e.g., using Apple Map to share location).

On the other hand, due to continuous advancements in technology at such a rapid pace, it is challenging to capture *all* the experiences of technology-facilitated abuse. The current survey was primarily about emotional and sexual violence resource awareness, with a small technology component. Therefore, we propose redesigning the survey with more nuanced questions capturing the technical ability, privacy mental model, and abuse experienced through technology at the forefront. Moreover, we plan to establish an open-source standard scale for technology-facilitated abuse with year-on-year checkpoint (similar to the TAR scale by Brown et al. [29]) that keeps up with the technological advancement of the

21st century. A standardized scale would be effective in understanding technological marginalization as more and more technology is used for social good or nefarious purposes.

ACKNOWLEDGMENTS

We acknowledge the Ho-Chunk Nation on whose ancestral lands we are grateful to work and live as a guest. We deeply respect the knowledge embedded in the Ho-Chunk's custodianship of Teejop (Madison) and recognize their continuing connection to land, water, and community here at the University of Wisconsin-Madison. Furthermore, we acknowledge funding from the Office for Victims of Crime, Office of Justice Programs, U.S. Department of Justice (Grant # 2020-V3-GX-0027 & 15POVC-23-GK-01414-NONF). The opinions, findings, and conclusions or recommendations expressed in this paper are those of the contributors and do not necessarily represent the official position or policies of the U.S. Department of Justice. Most importantly, we appreciate help from the members of Preventing Interpersonal Violence and Overcoming Trauma lab at UW-Madison and Sophie Stephenson for suggestions on research design. We thank the anonymous reviews that helped us shape the presentation of this paper.

REFERENCES

- [1] 2021. Statistics: National Statistics Domestic Violence Fact Sheet. <https://ncadv.org/STATISTICS>.
- [2] Zainab Agha, Karla Badillo-Urquiola, and Pamela J. Wisniewski. 2023. "Strike at the Root": Co-designing Real-Time Social Media Interventions for Adolescent Online Risk Prevention. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1 (April 2023), 1–32. <https://doi.org/10/gtmcnz>
- [3] Zainab Agha, Kelsey Miu, Sophia Piper, Jinkyung Park, and Pamela J. Wisniewski. 2023. Co-Designing User Personas and Risk Scenarios for Evaluating Adolescent Online Safety Interventions. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*. Association for Computing Machinery, New York, NY, USA, 249–253. <https://doi.org/10/gtmcpq>
- [4] Tousif Ahmed, Roberto Hoyle, Kay Connelly, David Crandall, and Apu Kapadia. 2015. Privacy Concerns and Behaviors of People with Visual Impairments. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems (CHI '15)*. Association for Computing Machinery, 3523–3532. <https://doi.org/10/ghfpdr>
- [5] Tousif Ahmed, Patrick Shaffer, Kay Connelly, David Crandall, and Apu Kapadia. 2016. Addressing Physical Safety, Security, and Privacy for People with Visual Impairments. 341–354. <https://www.usenix.org/conference/soups2016/technical-sessions/presentation/ahmed> (accessed 2023-09-11)
- [6] Taslima Akter, Bryan Dosono, Tousif Ahmed, Apu Kapadia, and Bryan Semaan. 2020. "I Am Uncomfortable Sharing What I Can't See": Privacy Concerns of the Visually Impaired with Camera Based Assistive Applications. 1929–1948. <https://www.usenix.org/conference/usenixsecurity20/presentation/akter> (accessed 2023-09-11)
- [7] J. Alemany, E. del Val, J. Alberola, and A. García-Fornes. 2019. Enhancing the Privacy Risk Awareness of Teenagers in Online Social Networks through Soft-Paternalism Mechanisms. *International Journal of Human-Computer Studies* 129 (Sept. 2019), 27–40. <https://doi.org/10/gpbrvh>
- [8] Eileen M. Alexy, Ann W. Burgess, Timothy Baker, and Shirley A. Smoyak. 2005. Perceptions of Cyberstalking Among College Students. 5, 3 (08 2005), 279. <https://doi.org/10/b2hn7w>
- [9] Majed Almansoori, Andrea Gallardo, Julio Poveda, Adil Ahmed, and Rahul Chatterjee. 2022. A Global Survey of Android Dual-Use Applications Used in Intimate Partner Surveillance. 2022, 4 (Oct. 2022), 120–139. <https://doi.org/10/gsn96x>
- [10] Donna L. Ansara and Michelle J. Hindin. 2010. Formal and Informal Help-Seeking Associated with Women's and Men's Experiences of Intimate Partner Violence in Canada. 70, 7 (April 2010), 1011–1018. <https://doi.org/10/dwrk3s>
- [11] Zahra Ashktorab and Jessica Vitak. 2016. Designing Cyberbullying Mitigation and Prevention Solutions through Participatory Design With Teenagers. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. Association for Computing Machinery, New York, NY, USA, 3895–3905. <https://doi.org/10.1145/2858036.2858548>
- [12] Avast. 2020. 51% increase in the use of online spying and stalking apps during lockdown — prnewswire.com. <https://www.prnewswire.com/news-releases/51-increase-in-the-use-of-online-spying-and-stalking-apps-during-lockdown-301090012.html>. [Accessed 05-Oct-2022].
- [13] Karla Badillo-Urquiola. 2020. A Social Ecological Approach to Empowering Foster Youth to Be Safer Online. In *Companion Publication of the 2020 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '20 Companion)*. Association for Computing Machinery, New York, NY, USA, 75–79. <https://doi.org/10.1145/3406865.3418365>
- [14] Karla Badillo-Urquiola, Xinru Page, and Pamela J. Wisniewski. 2019. Risk vs. Restriction: The Tension between Providing a Sense of Normalcy and Keeping Foster Teens Safe Online. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10/ghdmbb>
- [15] Karla Badillo-Urquiola, Zachary Shea, Zainab Agha, Irina Lediaeva, and Pamela Wisniewski. 2021. Conducting Risky Research with Teens: Co-designing for the Ethical Treatment and Protection of Adolescents. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW3 (Jan. 2021), 231:1–231:46. <https://doi.org/10/gtmcn3>
- [16] Jane Bailey, Asher Flynn, and Nicola Henry. 2021. Pandemics and Systemic Discrimination: Technology-Facilitated Violence and Abuse in an Era of COVID-19 and Antiracist Protest. In *The Emerald International Handbook of Technology Facilitated Violence and Abuse*, Jane Bailey, Asher Flynn, and Nicola Henry (Eds.). Emerald Publishing Limited, 787–797. <https://doi.org/10.1108/978-1-83982-848-520211057>
- [17] Victoria L. Banyard, Katie M. Edwards, Elizabeth A. Moschella, and Katherine M. Seavey. 2019. "Everybody's Really Close-Knit": Disconnections Between Helping Victims of Intimate Partner Violence and More General Helping in Rural Communities. 25, 3 (03 2019), 337–358. <https://doi.org/10/gnp9kv>
- [18] Chris Baraniuk. 2019. Stalkerware: The secret apps people use to spy on their partners — newscientist.com. <https://www.newscientist.com/article/mg24432572-600-stalkerware-the-secret-apps-people-use-to-spy-on-their-partners/>. [Accessed 05-Oct-2022].
- [19] Betty Jo Barrett and Melissa St. Pierre. 2011. Variations in Women's Help Seeking in Response to Intimate Partner Violence: Findings From a Canadian Population-Based Study. 17, 1 (Jan. 2011), 47–70. <https://doi.org/10/b4ftqq>
- [20] Rosanna Bellini. 2023. Paying the Price: When Intimate Partners Use Technology for Financial Harm. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 1–17. <https://doi.org/10/gr8rpz>
- [21] Rasika Bhalerao, Vaughn Hamilton, Allison McDonald, Elissa M. Redmiles, and Angelika Strohmayer. 2022. Ethical Practices for Security Research with At-Risk Populations. In *2022 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*. 546–553. <https://doi.org/10.1109/EuroSPW55150.2022.00065>
- [22] Katie Bishop. 2021. How 'Stalkerware' Technology Made It Easy For My Abusive Ex To Spy On Me — refinery29.com. <https://www.refinery29.com/en-gb/rise-of-stalkerware-tech>. [Accessed 05-Oct-2022].
- [23] Lindsay Blackwell, Jill Dimond, Sarita Schoenebeck, and Cliff Lampe. 2017. Classification and Its Consequences for Online Harassment: Design Insights from HeartMob. *Proceedings of the ACM on Human-Computer Interaction* 1, CSCW (Dec. 2017), 24:1–24:19. <https://doi.org/10/gqj8rq>
- [24] Lindsay Blackwell, Jean Hardy, Tawfiq Ammar, Tiffany Veinot, Cliff Lampe, and Sarita Schoenebeck. 2016. LGBT Parents and Social Media: Advocacy, Privacy, and Disclosure during Shifting Social Movements. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. Association for Computing Machinery, 610–622. <https://doi.org/10/grx6cc>
- [25] Warren J. Blumenfeld and R M Cooper. 2010. LGBT and Allied Youth Responses to Cyberbullying: Policy Implications. (2010).
- [26] Leanne Bowler, Eleanor Mattern, and Cory Knobel. 2014. Developing Design Interventions for Cyberbullying: A Narrative-Based Participatory Approach. *iConference 2014 Proceedings* (March 2014). <https://doi.org/10/gtmfk3>
- [27] Matthew J. Breiding, Kathleen C. Basile, Joanne Klevens, and Sharon G. Smith. 2017. Economic Insecurity and Intimate Partner and Sexual Violence Victimization. 53, 4 (Oct. 2017), 457–464. <https://doi.org/10/gckhbb>
- [28] Lynn Brewer. 2018. Domestic Violence and the LGBTQ Community. <https://ncadv.org/blog/posts/domestic-violence-and-the-lgbtq-community> (accessed 2023-05-17)
- [29] Cynthia Brown and Kelsey Hegarty. 2021. Development and Validation of the TAR Scale: A Measure of Technology-Facilitated Abuse in Relationships. 3 (Jan. 2021), 100059. <https://doi.org/10/gshf3d>
- [30] Chay Brown, Mandy Yap, Annick Thomassin, Minda Murray, and Eunice Yu. 2021. "Can I Just Share My Story?" Experiences of Technology-Facilitated Abuse among Aboriginal and Torres Strait Islander Women in Regional and Remote Australia. 5, 2 (Oct. 2021). <https://www.journalofglobalindigeneity.com/article/29716-can-i-just-share-my-story-experiences-of-technology-facilitated-abuse-among-aboriginal-and-torres-strait-islander-women-in-regional-and-remote-aust> (accessed 2023-06-19)
- [31] Megan Lindsay Brown, Judy Krysik, Walter LaMendola, Driшти Sinha, Lauren Reed, Megan Lindsay Brown, Judy Krysik, Walter LaMendola, Driшти Sinha, and Lauren Reed. 2020. Relationship Fluidity: The Changing Nature of Intimacy for Emerging Adult Women. <https://www.igi-global.com/gateway/chapter/www.igi-global.com/gateway/chapter/241032> (accessed 2023-06-27)
- [32] Sarah Jane Brubaker, Brittany Keegan, Xavier L. Guadalupe-Diaz, and Bre'Auna Beasley. 2017. Measuring and Reporting Campus Sexual Assault: Privilege and

- Exclusion in What We Know and What We Do. 11, 12 (Dec. 2017), e12543. <https://doi.org/10/ggfjsk>
- [33] Sloane C. Burke, Michele Wallen, Karen Vail-Smith, and David Knox. 2011. Using Technology to Control Intimate Partners: An Exploratory Study of College Undergraduates. 27, 3 (05 2011), 1162–1167. <https://doi.org/10/fmrhx2>
- [34] Nancy Chi Cantalupo and William C. Kidder. 2016. Mapping the Title IX Iceberg: Sexual Harassment (Mostly) in Graduate School by College Faculty. 66, 4 (2016), 850–881. <https://heinonline.org/HOL/P?h=hein.journals/jled66&i=857> (access ed2023-09-04)
- [35] Matthew Carrasco and Andrius Kerne. 2018. Queer Visibility: Supporting LGBT+ Selective Visibility on Social Media. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems (CHI '18)*. Association for Computing Machinery, 1–12. <https://doi.org/10/gmqk66>
- [36] CDC. 2021. Fast facts: Preventing intimate partner violence. 19 (2021), 2022.
- [37] Rose Ceccio, Sophie Stephenson, Danny Yuxing Huang, and Rahul Chatterjee. 2023. Sneaky Spy Devices and Defective Detectors: The Ecosystem of Intimate Partner Surveillance. (2023).
- [38] Rahul Chatterjee, Periwinkle Doerfler, Hadas Orgad, Sam Havron, Jackeline Palmer, Diana Freed, Karen Levy, Nicola Dell, Damon McCoy, and Thomas Ristenpart. 2018. The spyware used in intimate partner violence. In *2018 IEEE Symposium on Security and Privacy (SP)*. IEEE, 441–458.
- [39] Janet X. Chen, Allison McDonald, Yixin Zou, Emily Tseng, Kevin A Roundy, Acar Tamersoy, Florian Schaub, Thomas Ristenpart, and Nicola Dell. 2022. Trauma-Informed Computing: Towards Safer Technology Experiences for All. In *CHI Conference on Human Factors in Computing Systems*. ACM, 1–20. <https://doi.org/10.1145/3491102.3517475>
- [40] Ann L. Coker, Keith E Davis, Ileana Arias, Sujata Desai, Maureen Sanderson, Heather M Brandt, and Paige H Smith. 2002. Physical and mental health effects of intimate partner violence for men and women. 23, 4 (2002), 260–268.
- [41] Wendy Craig, Meyran Boniel-Nissim, Nathan King, Sophie D Walsh, Maartje Boer, Peter D Donnelly, Yossi Harel-Fisch, Marta Malinowska-Ciešlik, Margarida Gaspar de Matos, Alina Cosma, et al. 2020. Social media use and cyberbullying: A cross-national analysis of young people in 42 countries. 66, 6 (2020), S100–S108.
- [42] Kimberle Crenshaw. 1995. Mapping the margins. 3, 15 (1995), 357–383.
- [43] Walter S. DeKeseredy, Martin D. Schwartz, James Nolan, Nicholas Mastron, and Amanda Hall-Sanchez. 2019. Polyvictimization and the Continuum of Sexual Abuse at a College Campus: Does Negative Peer Support Increase the Likelihood of Multiple Victimization. 59, 2 (2019), 276–295. <https://doi.org/10/gshf25>
- [44] National Institute of Justice DOJ. 2023. Five Things About Violence Against American Indian and Alaska Native Women and Men. (2023).
- [45] Nathalie Dougé, Erik B Lehman, and Jennifer S McCall-Hosenfeld. 2014. Social support and employment status modify the effect of intimate partner violence on depression symptom severity in women: results from the 2006 behavioral risk factor surveillance system survey. 24, 4 (2014), e425–e434.
- [46] Suzie Dunn. 2021. Is It Actually Violence? Framing Technology-Facilitated Abuse as Violence. In *The Emerald International Handbook of Technology-Facilitated Violence and Abuse*, Jane Bailey, Asher Flynn, and Nicola Henry (Eds.). Emerald Publishing Limited, 25–45. <https://doi.org/10.1108/978-1-83982-848-520211002>
- [47] Brianna Dym, Jed R. Brubaker, Casey Fiesler, and Bryan Semaan. 2019. "Coming Out Okay": Community Narratives for LGBTQ Identity Recovery Work. 3 (Nov. 2019), 154:1–154:28. Issue CSCW. <https://doi.org/10/gkdmf9>
- [48] Katie M Edwards, Christina M Dardis, and Christine A Gidycz. 2012. Women's Disclosure of Dating Violence: A Mixed Methodological Study. 22, 4 (Nov. 2012), 507–517. <https://doi.org/10/f4fc4w>
- [49] Marcus Felson and Lawrence E Cohen. 1980. Human ecology and crime: A routine activity approach. 8 (1980), 389–406.
- [50] Bonnie S. Fisher, Francis T. Cullen, and Michael G. Turner. 2000. The Sexual Victimization of College Women. Research Report. <https://eric.ed.gov/?id=ED449712> (accessed2023-08-30)
- [51] Diana Freed, Natalie N. Bazarova, Sunny Consolvo, Eunice J Han, Patrick Gage Kelley, Kurt Thomas, and Dan Cosley. 2023. Understanding Digital-Safety Experiences of Youth in the U.S.. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10/gsw7s6>
- [52] Diana Freed, Sam Havron, Emily Tseng, Andrea Gallardo, Rahul Chatterjee, Thomas Ristenpart, and Nicola Dell. 2019. "Is my phone hacked?" Analyzing Clinical Computer Security Interventions with Survivors of Intimate Partner Violence. 3, CSCW (2019), 1–24.
- [53] Diana Freed, Jackeline Palmer, Diana Elizabeth Minchala, Karen Levy, Thomas Ristenpart, and Nicola Dell. 2017. Digital Technologies and Intimate Partner Violence: A Qualitative Analysis with Multiple Stakeholders. 1, CSCW (Dec. 2017), 1–22. <https://doi.org/10.1145/3134681>
- [54] Diana Freed, Jackeline Palmer, Diana Elizabeth Minchala, Karen Levy, Thomas Ristenpart, and Nicola Dell. 2017. Digital technologies and intimate partner violence: A qualitative analysis with multiple stakeholders. 1, CSCW (2017), 1–22.
- [55] FTC. 2021. How to Spot Extortion Scams on LGBTQ+ Dating Apps. <https://consumer.ftc.gov/consumer-alerts/2021/09/how-spot-extortion-scams-lgbtq-dating-apps>
- [56] Maite Garaigordobil and Vanesa Martínez-Valderrey. 2015. Effects of Cyberprogram 2.0 on "Face-to-Face" Bullying, Cyberbullying, and Empathy. *Psicothema* 27,1 (Feb. 2015), 45–51. <https://doi.org/10/gfdcrd>
- [57] Christine Geeng, Mike Harris, Elissa Redmiles, and Franziska Roesner. 2022. "Like Lesbians Walking the Perimeter": Experiences of {U.S.} {LGBTQ+} Folks With Online Security, Safety, and Privacy Advice. 305–322. <https://www.usenix.org/conference/usenixsecurity22/presentation/geeng> (accessed2023-02-08)
- [58] David Geller. 2020. An Increase in 'Stalkerware' is Posing Privacy Issues During COVID-19 – verisk.com. <https://www.verisk.com/insurance/covid-19/iso-insights/an-increase-in-stalkerware-is-posing-privacy-issues-during-the-pandemic/>. [Accessed 05-Oct-2022].
- [59] Mélissa Godin. 2020. How Domestic Abusers Have Exploited Technology During the Pandemic – time.com. <https://time.com/5922566/technology-domestic-abuse-coronavirus-pandemic/>. [Accessed 05-Oct-2022].
- [60] Oliver L. Haimson, Jed R. Brubaker, Lynn Dombrowski, and Gillian R. Hayes. 2015. Disclosure, Stress, and Support During Gender Transition on Facebook. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '15)*. Association for Computing Machinery, 1176–1190. <https://doi.org/10/gsnxrt>
- [61] Bridget Harris and Delanie Woodlock. 2022. 'You Can't Actually Escape It': Policing the Use of Technology in Domestic Violence in Rural Australia. 11, 1 (03 2022), 135–148. <https://doi.org/10/gr8rn2>
- [62] Tirion Elizabeth Havard and Michelle Lefevre. 2020. Beyond the Power and Control Wheel: How Abusive Men Manipulate Mobile Phone Technologies to Facilitate Coercive Control. 4, 2 (06 2020), 223–239. <https://doi.org/10/gnntq8>
- [63] Sam Havron, Diana Freed, Rahul Chatterjee, Damon McCoy, Nicola Dell, and Thomas Ristenpart. 2019. Clinical computer security for victims of intimate partner violence. In *28th {USENIX} Security Symposium ({USENIX} Security 19)*. 105–122.
- [64] Jennifer Hefner and Daniel Eisenberg. 2009. Social Support and Mental Health Among College Students. 79, 4 (2009), 491–499. <https://doi.org/10/d5tdbc>
- [65] Nicola Henry, Asher Flynn, and Anastasia Powell. 2018. Policing Image-Based Sexual Abuse: Stakeholder Perspectives. 19, 6 (Nov. 2018), 565–581. <https://doi.org/10/gn7zhz>
- [66] Nicola Henry, Asher Flynn, and Anastasia Powell. 2020. Technology-Facilitated Domestic and Sexual Violence: A Review. 26, 15-16 (Dec. 2020), 1828–1854. <https://doi.org/10/ghstg9>
- [67] Nicola Henry, Clare McGlynn, Asher Flynn, Kelly Johnson, Anastasia Powell, and Adrian J. Scott. 2020. *Image-Based Sexual Abuse: A Study on the Causes and Consequences of Non-consensual Nude or Sexual Imagery*. Routledge.
- [68] Nicola Henry and Anastasia Powell. 2018. Technology-Facilitated Sexual Violence: A Literature Review of Empirical Research. 19, 2 (04 2018), 195–208. <https://doi.org/10/gddz8g>
- [69] Nicola Henry and Alice Witt. 2021. Governing Image-Based Sexual Abuse: Digital Platform Policies, Tools, and Practices. In *The Emerald International Handbook of Technology-Facilitated Violence and Abuse*, Jane Bailey, Asher Flynn, and Nicola Henry (Eds.). Emerald Publishing Limited, 749–768. <https://doi.org/10.1108/978-1-83982-848-520211054>
- [70] Rebecca L. Heron and Maarten C. Eisma. 2021. Barriers and Facilitators of Disclosing Domestic Violence to the Healthcare Service: A Systematic Review of Qualitative Research. 29, 3 (May 2021), 612–630. <https://doi.org/10/gsbcc>
- [71] Jina Huh-Yoo, Afsaneh Razi, Diep N. Nguyen, Sampada Regmi, and Pamela J. Wisniewski. 2023. "Help Me:" Examining Youth's Private Pleas for Support and the Responses Received from Peers via Instagram Direct Messages. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10/gr6j4j>
- [72] Min-Wei Hung, Chien Wen (Tina) Yuan, Nanyi Bi, Yi-Chao Chen, Wan-Chen Lee, Ming-Chyi Huang, and Chuang-Wen You. 2022. To Use or Abuse: Opportunities and Difficulties in the Use of Multi-channel Support to Reduce Technology Abuse by Adolescents. 6, CSCW1 (April 2022), 125:1–125:27. <https://doi.org/10/gsq7k>
- [73] Naulsberry Jean Baptiste, Jinkyung Park, Neeraj Chatlani, Naima Samreen Ali, and Pamela J. Wisniewski. 2023. Teens on Tech: Using an Asynchronous Remote Community to Explore Adolescents' Online Safety Perspectives. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*. Association for Computing Machinery, New York, NY, USA, 45–49. <https://doi.org/10/gtmcpb>
- [74] Akiko Kamimura, Asha Parekh, and Lenora M Olson. 2013. Health indicators, social support, and intimate partner violence among women utilizing services at a community organization. 23, 3 (2013), e179–e185.
- [75] Jonathan J. Kandell. 1998. Internet Addiction on Campus: The Vulnerability of College Students. 1, 1 (01 1998), 11–17. <https://doi.org/10/db86jg>
- [76] Vanessa Kitzie. 2018. "I Pretended to Be a Boy on the Internet": Navigating Affordances and Constraints of Social Networking Sites and Search Engines for LGBTQ+ Identity Work. (07 2018). <https://doi.org/10.5210/fm.v23i7.9264>

- [77] L. B. Klein and Sandra L. Martin. 2021. Sexual Harassment of College and University Students: A Systematic Review. 22, 4 (Oct. 2021), 777–792. <https://doi.org/10/gsn7g9>
- [78] Shanti Kulkarni. 2019. Intersectional Trauma-Informed Intimate Partner Violence (IPV) Services: Narrowing the Gap between IPV Service Delivery and Survivor Needs. 34, 1 (Jan. 2019), 55–64. <https://doi.org/10.1007/s10896-018-0001-5>
- [79] Roxanne Leitão. 2021. Technology-Facilitated Intimate Partner Abuse: a qualitative analysis of data from online domestic abuse forums. 36, 3 (2021), 203–242.
- [80] Megan Lindsay and Judy Krysiak. 2012. Online Harassment Among College Students. 15, 5 (06 2012), 703–719. <https://doi.org/10/gndh2m>
- [81] Rebecca M. Loya. 2014. The Role of Sexual Violence in Creating and Maintaining Economic Insecurity Among Asset-Poor Women of Color. 20, 11 (Nov. 2014), 1299–1320. <https://doi.org/10.1177/1077801214552912>
- [82] Jeneile Luebke, Peninnah Kako, Alexa Lopez, Marin Schmitt, Anne Dressel, Kathryn Klein, and Lucy Mkandawire-Vahlmu. 2022. Barriers Faced by American Indian Women in Urban Wisconsin in Seeking Help Following an Experience of Intimate Partner Violence. (Oct. 2022), 10778012221132304. <https://doi.org/10/gsc3hk>
- [83] Minna Lyons and Gayle Brewer. 2022. Experiences of Intimate Partner Violence during Lockdown and the COVID-19 Pandemic. 37, 6 (Aug. 2022), 969–977. <https://doi.org/10.1007/s10896-021-00260-x>
- [84] Regan L Mandryk, Julian Frommel, Nitesh Goyal, Guo Freeman, Cliff Lampe, Sarah Vieweg, and Donghee Yvette Wohn. 2023. Combating Toxicity, Harassment, and Abuse in Online Social Spaces: A Workshop at CHI 2023. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. Association for Computing Machinery, 1–7. <https://doi.org/10/gsnxtc>
- [85] Alison Marganski and Lisa Melander. 2018. Intimate partner violence victimization in the cyber and real world: Examining the extent of cyber aggression experiences and its association with in-person dating violence. 33, 7 (2018), 1071–1095.
- [86] Alison Marganski and Lisa Melander. 2018. Intimate Partner Violence Victimization in the Cyber and Real World: Examining the Extent of Cyber Aggression Experiences and Its Association With In-Person Dating Violence. 33, 7 (04 2018), 1071–1095. <https://doi.org/10/gdd2qx>
- [87] Hiroaki Masaki, Kengo Shibata, Shui Hoshino, Takahiro Ishihama, Nagayuki Saito, and Koji Yatani. 2020. Exploring Nudge Designs to Help Adolescent SNS Users Avoid Privacy and Safety Threats. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10/gkcx3s>
- [88] Tara Matthews, Kathleen O'Leary, Anna Turner, Manya Sleeper, Jill Palzkill Woelfer, Martin Shelton, Cori Manthorne, Elizabeth F Churchill, and Sunny Consolvo. 2017. Stories from survivors: Privacy & security practices when coping with intimate partner abuse. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 2189–2201.
- [89] Tara Matthews, Kathleen O'Leary, Anna Turner, Manya Sleeper, Jill Palzkill Woelfer, Martin Shelton, Cori Manthorne, Elizabeth F. Churchill, and Sunny Consolvo. 2017. Stories from Survivors: Privacy & Security Practices When Coping with Intimate Partner Abuse. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, 2189–2201. <https://doi.org/10/gr8rpq>
- [90] Travis Mayberry, Erik-Oliver Blass, and Ellis Fenske. 2023. Blind My - An Improved Cryptographic Protocol to Prevent Stalking in Apple's Find My Network. 2023, 1 (01 2023), 85–97. <https://doi.org/10/gr8rpx>
- [91] Nora McDonald, Karla Badillo-Urquiola, Morgan G. Ames, Nicola Dell, Elizabeth Keneski, Manya Sleeper, and Pamela J. Wisniewski. 2020. Privacy and Power: Acknowledging the Importance of Privacy Research and Design for Vulnerable Populations. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10/gspbtz>
- [92] Nora McDonald and Andrea Forte. 2020. The Politics of Privacy Theories: Moving from Norms to Vulnerabilities. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, 1–14. <https://doi.org/10.1145/3313831.3376167>
- [93] Sarah McMahon and Kate Stepleton. 2018. Undergraduate Exposure to Messages About Campus Sexual Assault: Awareness of Campus Resources. 59, 1 (2018), 110–115. <https://doi.org/10/ggjfjm>
- [94] Lisa A Melander. 2010. College Students' Perceptions of Intimate Partner Cyber Harassment. (2010).
- [95] Terance D Miethe and Robert Frank Meier. 1994. *Crime and its social context: Toward an integrated theory of offenders, victims, and situations*. Suny Press.
- [96] Monica N. Modi, Sheallah Palmer, and Alicia Armstrong. 2014. The Role of Violence Against Women Act in Addressing Intimate Partner Violence: A Public Health Issue. 23, 3 (03 2014), 253–259. <https://doi.org/10/f54nz5>
- [97] Phoebe Moh, Pubali Datta, Noel Warford, Adam Bates, Nathan Malkin, and Michelle L. Mazurek. 2023. Characterizing Everyday Misuse of Smart Home Devices. In *2023 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2835–2849. <https://doi.org/10/gsnqv9>
- [98] Michelle L. Munro-Kramer, Alexandra C. Dulin, and Caroline Gaither. 2017. What Survivors Want: Understanding the Needs of Sexual Assault Survivors. 65, 5 (2017), 297–305. <https://doi.org/10.1080/07448481.2017.1312409>
- [99] Jordana N Navarro and Jana L. Jasinski. 2013. Why girls? Using routine activities theory to predict cyberbullying experiences between girls and boys. 23, 4 (2013), 286–303.
- [100] Borke Obada-Obieh, Yue Huang, Lucrezia Spagnolo, and Konstantin Beznosov. 2022. SoK: The Dual Nature of Technology in Sexual Abuse. In *2022 IEEE Symposium on Security and Privacy (SP)*. 2320–2343. <https://doi.org/10.1109/SP46214.2022.9833663>
- [101] Charlie Osborne. 2021. There's been a rise in stalkerware. And the tech abuse problem goes beyond smartphones — zdnet.com. <https://www.zdnet.com/article/theres-been-a-rise-in-stalkerware-and-the-tech-abuse-problem-goes-beyond-smartphones/>. [Accessed 05-Oct-2022].
- [102] Ellen Pence and Michael Paymar. 1993. *Education groups for men who batter: The Duluth model*. Springer Publishing Company.
- [103] Justin Petelka, Lucy Van Kleunen, Liam Albright, Elizabeth Murnane, Stephen Volda, and Jaime Snyder. 2020. Being (In)Visible: Privacy, Transparency, and Disclosure in the Self-Management of Bipolar Disorder. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, 1–14. <https://doi.org/10/gsqgrr>
- [104] Anthony T. Pinter, Morgan Klaus Scheuerman, and Jed R. Brubaker. 2021. Entering Doors, Evading Traps: Benefits and Risks of Visibility During Transgender Coming Outs. 4 (01 2021), 272:1–272:27. Issue CSCW3. <https://doi.org/10/gsnt2z>
- [105] John R. Porter, Kiley Sobel, Sarah E. Fox, Cynthia L. Bennett, and Julie A. Kientz. 2017. Filtered Out: Disability Disclosure Practices in Online Dating Communities. 1 (Dec. 2017), 1–13. Issue CSCW. <https://doi.org/10/gfgx4k>
- [106] Anastasia Powell and Nicola Henry. 2019. Technology-Facilitated Sexual Violence Victimization: Results From an Online Survey of Australian Adults. 34, 17 (09 2019), 3637–3665. <https://doi.org/10/gfttwz>
- [107] Anastasia Powell, Adrian Scott, Asher Flynn, and Nicola Henry. 2020. *Image-Based Sexual Abuse: An International Study of Victims and Perpetrators*. <https://doi.org/10.13140/RG.2.2.35166.59209>
- [108] Afsaneh Razi, Karla Badillo-Urquiola, and Pamela J. Wisniewski. 2020. Let's Talk about Sext: How Adolescents Seek Support and Advice about Their Online Sexual Experiences. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10/gtmcn6>
- [109] Lauren A. Reed, Richard M. Tolman, and L. Monique Ward. 2016. Snooping and Sexting: Digital Media as a Context for Dating Aggression and Abuse Among College Students. 22, 13 (Nov. 2016), 1556–1576. <https://doi.org/10/f875j7>
- [110] Eduardo Reis, Carla Moleiro, and Patricia Arriaga. 2023. Intimate Partner Violence Directed at Men: Experiences of Violence, Help-seeking, and Potential Gender Role Conflict Among Portuguese Men. (2023), 1–20.
- [111] Stuart Roe. 2017. "Family Support Would Have Been Like Amazing": LGBTQ Youth Experiences With Parental and Family Support. 25, 1 (Jan. 2017), 55–62. <https://doi.org/10/gmhgrm>
- [112] Michaela M. Rogers, Colleen Fisher, Parveen Ali, Peter Allmark, and Lisa Fontes. 2022. Technology-Facilitated Abuse in Intimate Relationships: A Scoping Review. (05 2022), 152483802210902. <https://doi.org/10/gr8rnw>
- [113] Wendy Roldan, Karla Badillo-Urquiola, Kiley Sobel, Kung Jin Lee, Pamela J. Wisniewski, June Ahn, Tamara Clegg, and Jason Yip. 2021. Justice-Centered Design Engagements with Children and Teens: What's at Stake, the Actions We Take, and the Commitments We Make. In *Proceedings of the 20th Annual ACM Interaction Design and Children Conference (IDC '21)*. Association for Computing Machinery, New York, NY, USA, 666–669. <https://doi.org/10.1145/3459990.3460515>
- [114] Jody M. Ross, Michelle Drouin, and Amanda Coupe. 2019. Sexting Coercion as a Component of Intimate Partner Polyvictimization. 34, 11 (06 2019), 2269–2291. <https://doi.org/10/gr8rpg>
- [115] Jody M Ross, Michelle Drouin, and Amanda Coupe. 2019. Sexting coercion as a component of intimate partner polyvictimization. 34, 11 (2019), 2269–2291.
- [116] Kevin A Roundy, Paula Barmaimon Mendelberg, Nicola Dell, Damon McCoy, Daniel Nissani, Thomas Ristenpart, and Acar Tamersoy. 2020. The many kinds of creepware used for interpersonal attacks. In *2020 IEEE Symposium on Security and Privacy (SP)*. IEEE, 626–643.
- [117] Nithya Sambasivan, Anna Batool, Nova Ahmed, Tara Matthews, Kurt Thomas, Laura Sanely Gaytán-Lugo, David Nemer, Elie Bursztin, Elizabeth Churchill, and Sunny Consolvo. 2019. "They Don't Leave Us Alone Anywhere We Go": Gender and Digital Abuse in South Asia. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, 1–14. <https://doi.org/10.1145/3290605.3300232>
- [118] Ari Schlesinger, W. Keith Edwards, and Rebecca E. Grinter. 2017. Intersectional HCI: Engaging Identity through Gender, Race, and Class. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, 5412–5427. <https://doi.org/10/cggw>
- [119] Ellen Simpson and Bryan Semaan. 2021. For You, or For "You"? Everyday LGBTQ+ Encounters with TikTok. 4 (01 2021), 252:1–252:34. Issue CSCW3. <https://doi.org/10/gsnxrm>

- [120] Julia Slupska. 2019. Safe at Home: Towards a Feminist Critique of Cybersecurity. <https://www.semanticscholar.org/paper/Safe-at-Home%3A-Towards-a-Feminist-Critique-of-Slupska/bf6752dcb067f3efa9f90f021f770ba70811077> (accessed 2023-02-15)
- [121] Julia Slupska and Megan Lindsay Brown. 2022. Aiding Intimate Violence Survivors in Lockdown: Lessons about Digital Security in the Covid-19 Pandemic. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22)*. Association for Computing Machinery, 1–5. <https://doi.org/10.1145/3491101.3503548>
- [122] Julia Slupska, Scarlet Dawson Dawson Duckworth, Linda Ma, and Gina Neff. 2021. Participatory Threat Modelling: Exploring Paths to Reconfigure Cybersecurity. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems (CHI EA '21)*. Association for Computing Machinery, 1–6. <https://doi.org/10.1145/3411763.3451731>
- [123] Julia Slupska and Angelika Strohmayer. 2022. Networks of Care: Tech Abuse Advocates' Digital Security Practices. In *31st USENIX Security Symposium (USENIX Security 22)*. 341–358.
- [124] Natalie J Sokoloff and Ida Dupont. 2005. Domestic violence at the intersections of race, class, and gender: Challenges and contributions to understanding violence against marginalized women in diverse communities. 11, 1 (2005), 38–64.
- [125] Marie Louise Juul Søndergaard, Gopinaath Kannabiran, Simran Chopra, Nadia Campo Woytuk, Dilrukshi Gamage, Ebtisam Alabdulqader, Heather McKinnon, Heike Winschiers-Theophilus, and Shaowen Bardzell. 2022. Feminist Voices about Ecological Issues in HCI. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems (CHI EA '22)*. Association for Computing Machinery, 1–7. <https://doi.org/10/gr5j6>
- [126] Sophie Stephenson, Majed Almansoori, Pardis Emami-Naeini, and Rahul Chatterjee. 2023. "It's the Equivalent of Feeling Like You're in Jail": Lessons from Firsthand and Secondhand Accounts of IoT-Enabled Intimate Partner Abuse. (2023).
- [127] Sophie Stephenson, Majed Almansoori, Pardis Emami-Naeini, Danny Yuxing Huang, and Rahul Chatterjee. 2023. Abuse Vectors: A Framework for Conceptualizing IoT-Enabled Interpersonal Abuse. (2023).
- [128] Kateryna M Sylaska and Katie M Edwards. 2014. Disclosure of intimate partner violence to informal social support network members: A review of the literature. 15, 1 (2014), 3–21.
- [129] NNEDV TechSafety. [n. d.]. DocuSAFE Documentation and Evidence Collection App. <https://www.techsafety.org/docusafe>
- [130] Kurt Thomas, Devdatta Akhawe, Michael Bailey, Dan Boneh, Elie Bursztein, Sunny Consolvo, Nicola Dell, Zakir Durumeric, Patrick Gage Kelley, Deepak Kumar, Damon McCoy, Sarah Meiklejohn, Thomas Ristenpart, and Gianluca Stringhini. 2021. SoK: Hate, Harassment, and the Changing Landscape of Online Abuse. In *2021 IEEE Symposium on Security and Privacy (SP)*. IEEE, 247–267. <https://doi.org/10.1109/SP40001.2021.00028>
- [131] Kristen Thomasen and Suzie Dunn. 2021. Reasonable Expectations of Privacy in an Era of Drones and Deepfakes: Expanding the Supreme Court of Canada's Decision in *R v Jarvis*. In *The Emerald International Handbook of Technology-Facilitated Violence and Abuse*, Jane Bailey, Asher Flynn, and Nicola Henry (Eds.). Emerald Publishing Limited, 555–576. <https://doi.org/10.1108/978-1-83982-848-520211040>
- [132] Joe Tidy. 2019. Stalkerware: The software that spies on your partner — bbc.com. <https://www.bbc.com/news/technology-50166147>. [Accessed 05-Oct-2022].
- [133] Emily Tseng, Diana Freed, Kristen Engel, Thomas Ristenpart, and Nicola Dell. 2021. A Digital Safety Dilemma: Analysis of Computer-Mediated Computer Security Interventions for Intimate Partner Violence During COVID-19. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, 1–17. <https://doi.org/10.1145/3411764.3445589>
- [134] Emily Tseng, Diana Freed, Kristen Engel, Thomas Ristenpart, and Nicola Dell. 2021. A Digital Safety Dilemma: Analysis of Computer-Mediated Computer Security Interventions for Intimate Partner Violence During COVID-19. 18, 22 (2021), 28–29.
- [135] Emily Tseng, Mehrnaz Sabet, Rosanna Bellini, Harkiran Kaur Sodhi, Thomas Ristenpart, and Nicola Dell. 2022. Care Infrastructures for Digital Security in Intimate Partner Violence. In *CHI Conference on Human Factors in Computing Systems*. ACM, 1–20. <https://doi.org/10.1145/3491102.3502038>
- [136] Kim Usher, Navjot Bhullar, Joanne Durkin, Naomi Gyamfi, and Debra Jackson. 2020. Family Violence and COVID-19: Increased Vulnerability and Reduced Options for Support. 29, 4 (Aug. 2020), 549–552. <https://doi.org/10.1111/inm.12735>
- [137] Judy A Van Wyk, Michael L Benson, Greer Litton Fox, and Alfred DeMaris. 2003. Detangling individual-, partner-, and community-level correlates of partner violence. 49, 3 (2003), 412–438.
- [138] Joseph C. Von Nessen and Erin Schubert. 2022. The Economic Impact of Domestic Violence In Milwaukee & Wisconsin 2021. <https://static1.squarespace.com/static/5d39f654dfc553000198b222/t/6345b7f7fc2276703a2bf19d/1665513464758/Economic+Impact+of+DV+FINAL.+10.11.2022.pdf> (accessed 2023-05-16)
- [139] Rachel J. Voth Schrag, Kristen E. Ravi, and Sarah R. Robinson. 2020. The Role of Social Support in the Link Between Economic Abuse and Economic Hardship. 35, 1 (Jan. 2020), 85–93. <https://doi.org/10.1007/s10896-018-0019-8>
- [140] Ari Ezra Waldman. 2021. Navigating Privacy on Gay-Oriented Mobile Dating Applications *. In *The Emerald International Handbook of Technology-Facilitated Violence and Abuse*, Jane Bailey, Asher Flynn, and Nicola Henry (Eds.). Emerald Publishing Limited, 369–381. <https://doi.org/10.1108/978-1-83982-848-520211027>
- [141] Ruolin Wang, Chun Yu, Xing-Dong Yang, Weijie He, and Yuanchun Shi. 2019. EarTouch: Facilitating Smartphone Use for Visually Impaired People in Mobile and Public Scenarios. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, 1–13. <https://doi.org/10/ghmkqs>
- [142] Noel Warford, Tara Matthews, Kaitlyn Yang, Omer Akgul, Sunny Consolvo, Patrick Gage Kelley, Nathan Malkin, Michelle L. Mazurek, Manyia Sleeper, and Kurt Thomas. 2022. SoK: A Framework for Unifying At-Risk User Research. In *2022 IEEE Symposium on Security and Privacy (SP)*. 2344–2360. <https://doi.org/10/gr8rp5>
- [143] Emily A. Waterman, Emily R. Dworkin, Christina M. Dardis, Sarah E. Ullman, Katie M. Edwards, and Lindsey M. Rodriguez. 2021. Exploring the Association Between Anticipated and Actual Responses to Disclosures of Intimate Partner Violence and Sexual Assault. 38, 4 (April 2021), 1131–1151. <https://doi.org/10/g99bcb>
- [144] Jennifer M. Watson, Michele Cascardi, Sarah Avery-Leaf, and K. Daniel O'Leary. 2001. High School Students' Responses to Dating Aggression. 16, 3 (Jan. 2001), 339–348. <https://doi.org/10/g99bb9>
- [145] Leila Wood, Rachel Voth Schrag, Dixie Hairston, and Cynthia Jones. 2021. Exploring Advocacy Practices for Interpersonal Violence Survivors on College Campuses: Approaches and Key Factors. 11, 1 (01 2021), 28–39. <https://doi.org/10/gmvf9x>
- [146] Delanie Woodlock. 2017. The Abuse of Technology in Domestic Violence and Stalking. 23, 5 (April 2017), 584–602. <https://doi.org/10.1177/1077801216646277>
- [147] Delanie Woodlock and Bridget Harris. 2022. 'You Have to Be Really Careful': Technology and the Abuse of Women with Intellectual and Cognitive Disabilities. (08 2022), 1–21. <https://doi.org/10/gscxq6>
- [148] Delanie Woodlock, Michael Salter, Molly Dragiewicz, and Bridget Harris. 2023. "Living in the Darkness": Technology-Facilitated Coercive Control, Disenfranchised Grief, and Institutional Betrayal. 29, 5 (04 2023), 987–1004. <https://doi.org/10/gscxq7>
- [149] Emily M Wright. 2015. The relationship between social support and intimate partner violence in neighborhood context. 61, 10 (2015), 1333–1359.
- [150] Michelle Ybarra, Myeshia Price-Feeney, Amanda Lenhart, and Kathryn Zickuhr. 2017. Intimate Partner Digital Abuse. (2017).
- [151] G. D. Zimet, S. S. Powell, G. K. Farley, S. Werkman, and K. A. Berkoff. 1990. Psychometric Characteristics of the Multidimensional Scale of Perceived Social Support. 55, 3-4 (1990), 610–617. <https://doi.org/10/bjdp5q>

A APPENDIX

Table 4: Regression Results for Support-Seeking. The number represents the Odds Ratio (OR) and 95% confidence intervals in brackets.

	No Support taken	Informal Support
Year [Graduate]	1.117 (0.556, 1.678)	0.855 (0.422, 1.288)
Ethnicity [Hispanic]	1.149 (0.498, 1.799)	0.825 (0.354, 1.297)
Relationship [Married]	1.396 (0.083, 2.708)	0.539 (0.012, 1.066)
Relationship [Monogamous]	0.847 (0.505, 1.188)	1.226 (0.729, 1.723)
Relationship [Non-Exclusive]	0.720 (0.377, 1.064)	1.372 (0.716, 2.029)
Low Income [Yes]	1.012 (0.643, 1.382)	1.023 (0.648, 1.399)
First Generation [Yes]	0.822 (0.450, 1.193)	1.241 (0.676, 1.807)
Disabled [Yes]	1.024 (0.560, 1.487)	0.994 (0.543, 1.446)
Gender [Cis-Women]	0.355*** (0.148, 0.562)	2.964*** (1.217, 4.712)
Gender [LGBTQ+]	0.441** (0.157, 0.725)	2.432*** (0.848, 4.016)
Race [BIPOC]	0.552 (0.106, 0.998)	1.601 (0.280, 2.922)
Gender [Cis-Women] & Race [BIPOC]	1.289 (-0.014, 2.592)	0.838 (-0.021, 1.697)
Gender [LGBTQ+] & Race [BIPOC]	1.356 (-0.035, 2.747)	0.837 (-0.034, 1.708)
Constant	2.994*** (1.276, 4.712)	0.307*** (0.128, 0.486)
Observations	545	545
Log Likelihood	-365.148	-361.984
Akaike Inf. Crit.	758.297	751.967

P-values: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$